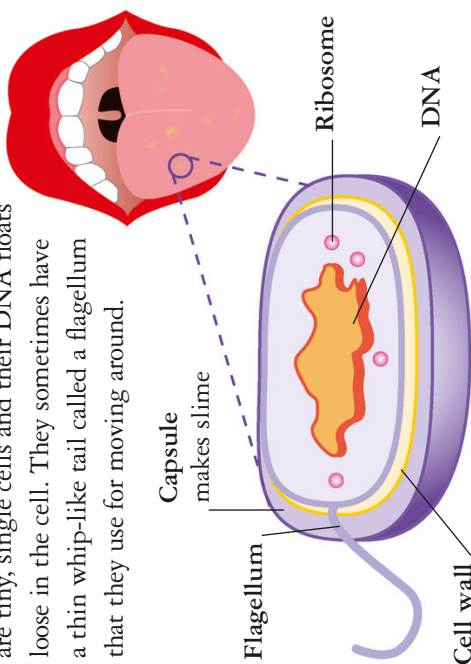


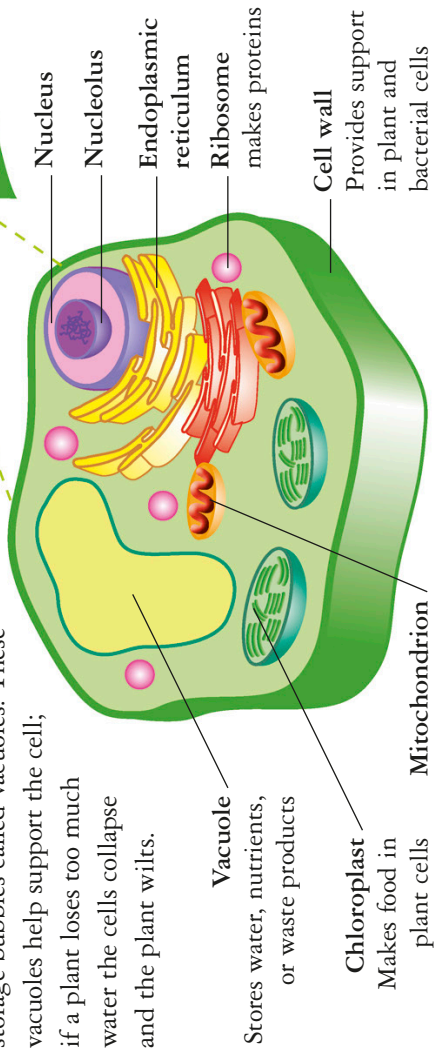
Prokaryotes

Prokaryotes, such as bacteria, are the oldest type of organism on Earth and have a simple cell structure. They are tiny, single cells and their DNA floats loose in the cell. They sometimes have a thin whip-like tail called a flagellum that they use for moving around.



Plant cells

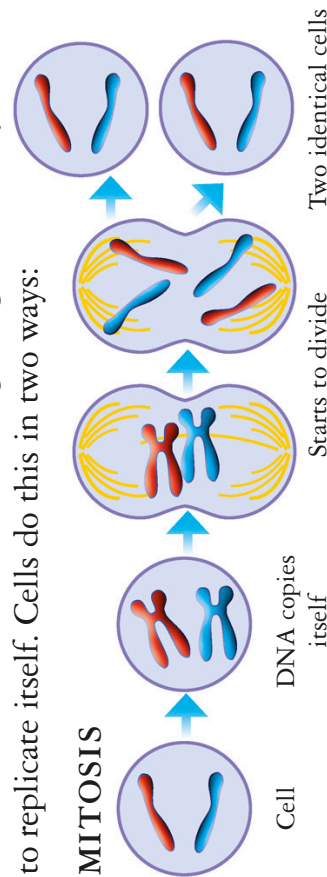
These are eukaryotes, but they differ from animal cells in having a stiff cell wall made of cellulose. Plant cells also contain green organelles called chloroplasts and large water storage bubbles called vacuoles. These vacuoles help support the cell; if a plant loses too much water the cells collapse and the plant wilts.



MAKING MORE CELLS

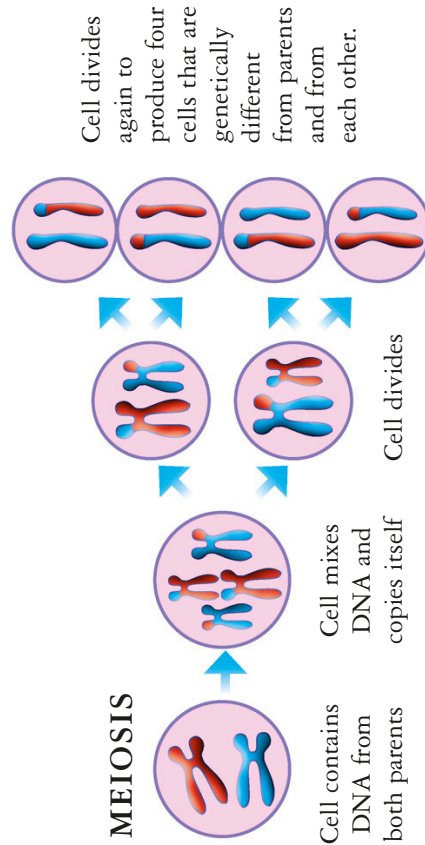
One of the characteristics of a living thing is its ability to replicate itself. Cells do this in two ways:

MITOSIS



Single-celled eukaryotes use a process called **mitosis**. The DNA of the cell makes a copy of itself and then lines itself up across the middle of the cell. The cell then begins to split in two so that each half of the cell has a copy of the DNA molecule. Somehow the dividing cell arranges so that each half of the cell has enough of the key organelles to function. This process results in two genetically identical cells.

MEIOSIS



More advanced eukaryotes reproduce by **meiosis** to produce egg and sperm cells. Just before it splits, the cell mixes the DNA so that it has sections from both parents. Next, the cell divides to produce four cells, each containing a single set of DNA. Each cell then has to fuse with another during sexual reproduction to get a full set of DNA. At this stage the fused cell can copy itself by mitosis, making more cells so that the organism can grow.